

Supplementary Document: Summary of Revisions

Submitted alongside our IJRR submission for transparency

Title: Physically-Feasible Reactive Synthesis for Terrain-Adaptive Locomotion

**Submitted to:
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**Prior review history (incorporated into this submission):
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Summary of Revisions

Dear IJRR Editors and Reviewers,

An earlier version of this manuscript received detailed feedback from three reviewers at IEEE Transactions on Robotics (T-RO Manuscript 25-1489), with the editor concluding that the manuscript at that time did not yet meet the bar for “new results of substantive research significance and impact beyond the previous papers.” We took the reviewers’ technical concerns seriously and have made substantial efforts in this revision to strengthen the manuscript along every dimension they identified. This document is provided as supplementary material to make the revision history transparent and to help the IJRR review process by showing precisely how each prior comment was addressed.

At the structural level, the contribution narrative has been reworked so that the framework’s complete technical scope is visible end-to-end. The conference-introduced scalability machinery (the high-level manager and symbolic repair with dynamic feasibility checks) is now presented alongside—and at the same level of detail as—the journal-specific mechanisms developed for this article: delay-aware coordination of the strategy automaton with the tracking controller, kinematic-feasibility re-targeting in the online MICP, and collision-region selection with swing-foot clearance constraints. Each of the journal-specific mechanisms is discussed in the introduction with explicit motivation for why it closes a concrete deployment gap, rather than presented as a standalone novelty bullet.

The major changes in this submission are:

1. Contribution narrative. The introduction now lists four contributions: (i) the integrated reactive-synthesis-and-MICP framework; (ii) scalable offline synthesis via the high-level manager and symbolic repair, with concrete 80.9–90.1% state-space-reduction and 71.6–97.6% MICP-solve-reduction numbers; (iii) the online-execution module with new mechanisms introduced in this article—kinematic-feasibility re-targeting, delay-aware coordination between the strategy automaton and the tracking controller, and collision-region selection in the online MICP; (iv) comprehensive empirical validation including pure-MIP and heuristic baselines, hardware deployment on two robot platforms, and a preliminary yaw-extension feasibility study. The introduction also includes a dedicated paragraph that motivates each of the journal-specific online-execution mechanisms with explicit discussion of the deployment gap it closes.

2. Online-execution clarity. Section 6 (online execution) was the most-criticized section in the prior round. The collision-avoidance subsection has been rewritten to explicitly distinguish between collision-free region selection and swing-foot clearance, name which robot parts are checked, and acknowledge that perception is out of scope. A new short subsection explains why symbolic-continuous interaction operates at the transition timescale rather than per-NMPC-tick.

3. Quantitative presentation overhaul, with comprehensive results for symbolic repair. Several reviewers asked for more quantitative evidence on how often and how expensively repair is performed; the IJRR submission addresses this at both the offline and runtime levels. Table 3 (*Offline Synthesis and Repair Time*) reports per-scenario success/total state-pair counts, original/new/total skill counts, symbolic-repair wall-clock times, and gait-fixed/gait-free feasibility-check times: across the four scenarios, repair successfully resolves 93.4% of all terrain-request state pairs, and the number of newly discovered skills is reduced by 71.6–97.6% relative to exhaustive enumeration. A new *Runtime Repair Case Study* subsection (Sec. 8.6) reports concrete timing for the two run-time scenarios—unforeseen terrain (12.36 s total, 0.18 s symbolic repair plus 12.18 s gait-free MICP) and online MICP solving failure (0.11 s and 0.07 s for the 7-type and 14-type rebar

cases). The introduction’s scalability-motivation paragraphs cite the manager’s 80.9–90.1% symbolic state-space reduction directly. On the implementation side, the results section now declares the solver (Gurobi 9), states the hardware platform (i9-13900K, 64 GB RAM), defines the pure-MIP baseline precisely, and adds a std-dev column to the hardware MICP-solve-time table.

4. Strengthened technical soundness of the newly proposed mechanisms via a new benchmark and explicit ablations. The Pure-MIP benchmark subsection is substantially expanded. Fig. 11 now reports per-MIP solve time across four terrains and five planning horizons as box plots aggregated over 10 seeded waypoints per scenario. A new end-to-end benchmark on the dense-rebar scenario (Table 5) compares the proposed planner against pure-MIP $H=2$ s and $H=3$ s baselines and against two ablations of the newly proposed mechanisms (disabling the MIP feasibility check; disabling delay-aware coordination), with success rate, mean traversal time, and per-module computational cost reported per trial. The proposed planner matches the pure-MIP $H=3$ s success rate (90%) at $\sim 75\%$ of its traversal time and $< 50\%$ of its MIP-transition cost; disabling the feasibility check drops the success rate from 90% to 60%, and disabling delay-aware coordination inflates the mean traversal time from 18.5 s to 23.4 s—directly demonstrating that each newly proposed mechanism is necessary for the reported performance. A representative qualitative comparison is shown in Fig. 12.

5. Calibrated claims and honest limitations. We tone down “dynamically changing environments,” “dynamic locomotion,” and “safety guarantees” wherever the experimental evidence does not warrant the strongest reading. We disclose the angular-momentum simplification of the MICP dynamics. The yaw extension is demoted to a preliminary feasibility study with explicit caveats. The discussion section is expanded with paragraphs on perception uncertainty, stronger future-baseline comparisons (Winkler 2018 phase-based parameterization and perceptive RL controllers), and bipedal-extension considerations.

All revisions to the manuscript are highlighted in blue text to differentiate them from prior content. In the per-reviewer sections that follow, the original reviewer comments are boxed and our responses are detailed underneath, with revised passages quoted in blue italic for the reviewer’s convenience. Unless otherwise noted, references to sections, figures, and equations in our responses point to the IJRR manuscript; references inside the boxed reviewer comments point to the prior T-RO version of the manuscript.

Sincerely,
Ziyi Zhou, Qian Meng, Hadas Kress-Gazit, and Ye Zhao

Acknowledgement Of Prior Editorial Feedback

Editor’s Overall Comments (T-RO 25-1489)

All reviewers raise a major concern that the current manuscript is a direct extension of [zhou2025physically], published in 2025 in the IROS proceedings. While this submission essentially presents additional hardware experiments, it reuses the very same materials and approaches. . . . Unfortunately, in the current state, the submission does not provide new results of substantive research significance and impact beyond the previous papers.

Response

We thank the editor for the candid feedback. The shared concern across the three reviewers—that the technical contribution of the journal extension required stronger justification—was fair, and we have made substantial efforts to address it in this revision. The journal-specific mechanisms (delay-aware coordination, kinematic-feasibility re-targeting, and collision-region selection in the online MICP) are now described in the introduction with explicit motivation for why each closes a concrete deployment gap, and the contribution narrative has been reworked so that the conference-introduced scalability machinery and the new mechanisms are presented end-to-end at the same level of detail. Most importantly, we have **strengthened the technical soundness of the newly proposed mechanisms with new quantitative evidence and explicit ablations**: the Pure-MIP benchmark subsection now includes (i) a per-MIP solve-time scaling study over 10 seeded waypoints per scenario and (ii) a new end-to-end benchmark on the dense-rebar scenario (Table 5) reporting success rate, mean traversal time, and per-module computational cost, with explicit ablation rows for the MIP feasibility check (disabling it drops success from 90% to 60%) and delay-aware coordination (disabling it inflates traversal time from 18.5s to 23.4s). We have also expanded the rest of the technical and quantitative content along the dimensions the reviewers requested: a clearer treatment of obstacle handling at the symbolic and MICP levels, per-trial standard deviations on the hardware MICP-solve-time table, an offline-synthesis-and-repair-time table with concrete success rates and skill-discovery counts, a runtime-repair case study, calibrated dynamic-locomotion and safety-guarantee claims, an angular-momentum disclosure on the MICP dynamics, and a new *Reactivity and Update Frequencies* discussion subsection. The contribution list now reads:

1. The integrated reactive-synthesis-and-MICP framework for terrain-adaptive locomotion.
2. Scalable offline synthesis via the high-level manager and symbolic repair with dynamic feasibility checks (with concrete 80.9–90.1% state-space-reduction and 71.6–97.6% MICP-solve-reduction numbers).
3. The online-execution module bridging offline synthesis and real-world deployment, with new mechanisms introduced in this article: kinematic-feasibility re-targeting, delay-aware coordination between the strategy automaton and the tracking controller, and collision-region selection in the online MICP.
4. Comprehensive empirical validation: simulation across four scenarios, two baseline comparisons, hardware deployment on two robot platforms, and a preliminary yaw-extension feasi-

bility study.

The detailed reviewer-by-reviewer responses follow on the next pages.

Response To Reviewer #4

Overall Comments

The paper proposes to reduce the combinatorial complexity of Mixed Integer approaches for locomotion planning by adding a layer of abstractions in the planning to limit the planning horizon while considering long-term objectives ... The approach is interesting, and the results qualitatively strong. However, the novelty of the paper (wrt to [zhou2025physically]) focuses on hardware experiments and reactivity, but there are insufficient details on this aspect to get a clear idea of what is implemented by the authors. The quantitative analysis does not really allow comparing methods with each other either, and would benefit from rewriting.

Response

We thank the reviewer for the thoughtful review and for engaging with the technical content. The IJRR submission addresses the contribution-framing concern by restating the conference contributions (the integrated framework, the manager, and symbolic repair) as bullets 1 and 2 of the contribution list and presenting the journal extensions (re-targeting, delay-aware coordination, collision-region selection, hardware, baselines, yaw study) as additions on top. The technical-clarity and quantitative-presentation concerns are addressed item by item below.

Reviewer Comment 1: Angular Momentum

While the authors claim that the solutions are dynamically consistent, it is essential to be clear about the fact that angular momentum is completely neglected by the approach. This is a well-known limitation of MIP solvers, making their use tricky for any form of truly dynamic motions, or for applications to humanoid robots. This is not clearly stated, and the claims of feasibility should be toned down.

Response

We agree and have added an explicit disclosure in Sec. 5 immediately after Eq. (3):

We note that the rotational dynamics adopted in Eq. (3) are intentionally simplified: the rotational equation of motion is replaced by a double integrator on the Euler angles, which keeps the constraint convex but neglects the true angular-momentum dynamics. This makes the formulation tractable for online deployment, but it limits applicability to highly dynamic motions such as flips, aggressive yaw maneuvers at high speed, or platforms where angular-momentum management is critical (e.g., humanoids). Future work will explore convex relaxations of the full angular-momentum dynamics to expand the range of feasible motions while maintaining online tractability.

We also tone down “dynamic locomotion” / “dynamically consistent” framings throughout the abstract, introduction, and hardware sections, and explicitly link the angular-momentum limitation to the yaw extension caveats (Sec. 8.8) and the bipedal-extension discussion (Sec. 10).

Reviewer Comment 2: Terrain Segmentation Module

“the online gait-fixed MICP is executed given accurate terrain information from a terrain segmentation module”. We definitely need more information on that module!

Response

We have added an explicit perception-scope statement in Sec. 6:

We emphasize that perception is intentionally treated as out of scope in this paper. We assume an upstream segmentation pipeline (e.g., elevation-mapping with plane segmentation (Miki et al., 2022b; Fankhauser et al., 2018)) provides labeled convex polygons; in our hardware experiments we use motion-capture-aided manual labeling of polygons, which is sufficient to validate the planning and control framework but does not represent a fully autonomous perception stack.

We further added a paragraph in the discussion (Sec. 10) on how perception uncertainty would propagate through the framework and the partial fallback that runtime symbolic repair provides.

Reviewer Comment 3: Collision-Free Regions Section

The section introduces collision-free regions for obstacle avoidance, but the paper does not specify how obstacles are detected or how these regions are constructed from sensor or terrain data. It is also unclear which parts of the robot are considered for collision checking. . . . The description suggests that collision-free regions are selected rather than explicit collision avoidance being enforced, yet the section then discusses swing-foot height parametrization, which implies stepping over obstacles—something that is not demonstrated in the results. Overall, this section is confusing and would benefit from substantial clarification.

Response

We rewrote the collision-avoidance subsection to make explicit that obstacle handling in our framework happens at *two distinct levels*, and to clarify which scenarios exercise which mechanism. Large obstacles that occupy entire abstracted cells are handled at the symbolic level—the affected cell is classified as the *obstacle* terrain class, and an obstacle that appears at runtime triggers symbolic re-synthesis to find a detour. Smaller obstacles that the swing foot must clear within a single transition are handled at the MICP level via the binary collision-region variables $\mathbf{H}_{s,w,j}$ together with a swing-foot height threshold h_{swing} . We also state which robot parts are checked, clarify how obstacles are labeled (manually annotated via mocap in the hardware experiment), and we no longer claim full-body mesh-level collision checking. The revised text reads:

Symbolic-level obstacle avoidance: when an obstacle is large enough that an entire abstracted cell becomes unsteppable, it is classified as the obstacle terrain class. The symbolic specification rules out transitions into such cells, and if an obstacle appears at runtime—as in the rebar-with-obstacle hardware experiment, where the obstacle is detected via motion capture and added to the local abstraction online—runtime symbolic re-synthesis finds a detour around it. No collision-region variables are added to the online MICP in this case.

MICP-level collision-free region selection: for smaller obstacles that the swing foot must clear within a single symbolic transition, we encode collision avoidance directly in the online MICP.

Binary variables $\mathbf{H}_{s,w,j}$ constrain the swing-foot trajectory of foot j to lie inside one of S pre-labeled collision-free convex polytopes at each time step. These polytopes come from the same labeled-polygon segmentation pipeline as the steppable terrain, with a separate label set distinguishing free-space polytopes from steppable polygons. In simulation, this mechanism is exercised in the Unstructured-8 case with collision avoidance enabled (Table 4), where the binary-variable count rises to 256.

The collision-free region selection in Eq. (29) and the steppable-region selection in Eq. (6) handle different geometric primitives: the steppable-region constraint forces stance-foot placement onto a labeled steppable terrain polygon at the first stance time step, while the collision-free region constraint keeps the swing-foot trajectory inside a free-space polytope across all time steps within a swing phase. The two are complementary, not redundant.

A separate user-tunable swing-foot height threshold h_{swing} enforces clearance margins for cases such as jumping onto a higher platform, where the collision-free region selection alone would require fine time-step discretization of the binary variables and would tend to yield trajectories that only marginally clear the terrain.

Reviewer Comment 4: Perception Capabilities Overstated

... the presentation implicitly suggests that identifying convex collision-free regions online is straightforward. In general, when terrain surfaces do not exhibit such regular structure, this is not the case. ... The authors should either clarify the perception tools and assumptions used to obtain these regions or tone down the emphasis on online perception capabilities.

Response

We took the second option: we explicitly tone down the perception claims and label perception as out-of-scope. The added text (quoted in the response to Comment 2 above) makes clear that obtaining convex collision-free regions in unstructured environments is non-trivial, that we make no claim of solving this problem, and that our hardware uses motion-capture-aided manual labeling. The discussion section now also explicitly lists robust perception integration as future work.

Reviewer Comment 5: Yaw Orientation Section

The orientation change is not demonstrated in the companion video, only in Figure 10. ... The authors claim in the title of the section that they demonstrate generalizability, but the work is very preliminary and nothing is really demonstrated. I would suggest removing this section entirely, or being clearer on the results obtained.

Response

We took the second of the reviewer’s two suggestions (“be clearer”). The section is now titled *Preliminary Study: Yaw Command Extension* (Sec. 8.8) and opens with explicit caveats:

This subsection demonstrates that the proposed framework can be extended to non-zero yaw waypoints in principle, but we report the result as a preliminary feasibility study only. We do

not claim a full demonstration of generalizability. Specifically: (i) the symbolic abstraction grows combinatorially when yaw is added as a discretized input ...; (ii) the angular-momentum coupling neglected by the convex-dynamics simplification becomes more relevant once yaw is treated as a planned variable, especially for fast turns; (iii) we report simulation results only—no hardware deployment of the yaw extension.

We retained the section because it shows the symbolic-decomposition machinery (preconditions/postconditions, manager, repair) generalizes to additional state dimensions, which is informative for readers considering the framework for other platforms or motion modes.

Reviewer Comment 6: MIP Solver

How is the MIP implemented? With which solver?
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Response

We added a new *Implementation Details* subsection at the start of the results section:

All MICP problems—including the offline gait-free MICP, the online gait-fixed MICP, the kinematic-feasibility re-targeting MICP, and the pure-MIP benchmark—are solved using Gurobi 9. Experiments are run on a desktop workstation equipped with an Intel Core i9-13900K (24 cores) and 64 GB of RAM.

Reviewer Comment 7: Per-Module Timing Decomposition

Table III and Fig. 9 are the only quantitative results regarding the comparisons of the MIP and the proposed framework, but they do not compare the same scenarios nor the same thing. ... It would clarify things to compare the global running time of the proposed on each scenario, with a proper decomposition of the time spent in each module. Obtaining the actual computation times upon reading the paper proves challenging because the information is disseminated.

Response

We agree that timing data was scattered. The biggest change in this revision is the new **end-to-end benchmark** on the dense-rebar scenario (Sec. 8, Table 5), which consolidates all the relevant quantities into one table: *success rate*, *mean traversal time*, and *mean per-module active computation per trial*—broken down into MIP transition solving, the pre-flight MIP feasibility check, reactive synthesis (strategy query plus reload), and runtime repair. Each method is dispatched against 10 identically seeded random waypoints, with random rebar removals between trials, so any difference reflects the planner alone.

The proposed planner attains 90% success at 18.5s mean traversal time and a per-trial decomposition of 5.26s MIP transition solving, 0.19s MIP feasibility check, 0.29s reactive synthesis, and 0.03s runtime repair. The pure-MIP $H=3$ s baseline matches the 90% success rate but at 24.7s traversal and 11.54s MIP transition cost—roughly $2\times$ the MIP cost for $\sim 33\%$ longer traversal—and the $H=2$ s pure-MIP baseline drops to 60%. The pure-MIP definition (now precisely stated in Sec. 8) is shared with our online gait-fixed MICP except for the absence of symbolic decomposition and the horizon-induced expansion of the local planning region:

The pure-MIP baseline uses the same convex relaxations of the dynamics, the same linearized friction-cone and fixed-Jacobian kinematic constraints, and the same cost weights as our online gait-fixed MICP. The only differences are (i) the planner solves a single long-horizon gait-fixed MICP without symbolic-level decomposition, and (ii) the local planning region is expanded to encompass all terrain polygons within the horizon-induced reachable distance.

To make the per-MIP comparison directly readable: the $H=2$ s horizon in Fig. 11 is configured with a planning region of two cells (i.e., two consecutive symbolic transitions in our framework), with each additional second of horizon adding half a cell. We also augmented the hardware MICP-solve-time table with a std-dev column for variability. Offline-synthesis and repair statistics are reported in Table 3; the new *Runtime Repair Case Study* subsection (Sec. 8.6) covers the runtime side. Together these address both the per-module-timing breakdown and the apples-to-apples comparison the reviewer asked for.

Reviewer Comment 8: Quantitative Repair Data + Short-Horizon MPC Comparison

There are no quantitative data on the amount of repair necessary, etc. . . . What if a short-horizon MPC was used to repeatedly solve the proposed scenarios in a receding-horizon fashion, with a fixed gait for the rebar scenario? . . . Without such comparisons, the interest of using a long-term planner is not demonstrated.

Response

We have added quantitative repair data at two levels.

Offline repair statistics. Table 3 (*Offline Synthesis and Repair Time*) reports, for each scenario–grid-size combination, the number of successful versus total terrain/request state pairs, the original/new/total skill counts, the symbolic-repair wall-clock time, and the gait-fixed/gait-free MICP feasibility-check times. Across the four scenarios, repair successfully resolves 93.4% of the terrain–request state pairs, and the number of newly discovered skills is reduced by 71.6–97.6% relative to exhaustive enumeration over all possible skills—directly answering the reviewer’s request for quantitative “amount of repair necessary” data.

Runtime repair case studies. A new *Runtime Repair Case Study* subsection (Sec. 8.6) covers two run-time scenarios with concrete timing numbers:

(Unforeseen Terrain States) When the robot reaches a new terrain state that contains a gap excluded from offline synthesis, runtime repair is triggered. The runtime repair takes 12.36 s in total, of which the symbolic-repair stage accounts for 0.18 s and the gait-free MICP for the new leaping gait accounts for 12.18 s.

(Solving Failure) Re-synthesis triggered by online MICP solving failures takes 0.11 s and 0.07 s in the 7-type and 14-type rebar cases, respectively, without invoking full runtime repair because the existing skill set already suffices.

Receding-horizon MPC comparison. The new end-to-end benchmark on the dense-rebar scenario (Sec. 8, Table 5) directly addresses this question. With a fixed gait and convex relaxations identical to ours, repeatedly solving a short-horizon MIP in a receding-horizon fashion is exactly our pure-MIP $H=2$ s and $H=3$ s configurations. The data show that:

- *Short horizon is insufficient.* The pure-MIP $H=2$ s baseline (planning footprint matched to one of our symbolic transitions) attains only 60% success vs. 90% for the proposed planner.

- *Longer horizon recovers success but is expensive.* The pure-MIP $H=3$ s baseline matches our 90% success rate but at 24.7 s mean traversal time vs. our 18.5 s, and at 11.54 s MIP transition cost per trial vs. our 5.26 s.

The qualitative failure mode is illustrated in Fig. 12: when the short-horizon pure-MIP planner cannot find a feasible foothold pattern inside its local window, it has no mechanism to deliberate at the symbolic level about which neighboring cell to attempt next, so it stalls; the proposed planner instead routes through a different sequence of cells via the symbolic strategy. The related question of how the proposed framework would scale to higher-frequency MICP updates is addressed in the new *Reactivity and Update Frequencies* subsection of the discussion (Sec. 10), which notes that more frequent MICP updates are easily adaptable in our framework via the delay-aware coordination scheme.

Reviewer Comment 9: Symbolic Repair Introduced Too Late + Eq. 1 Notation

Symbolic repairing is repeatedly referenced to but only introduced in Section VI.
 ... Eq. 1 does not specify that $\mathcal{D}$ is the system dynamics.

Response

For symbolic repair, we added an informal forward-definition at the point where the term first appears in the *Specifications* subsection of Sec. 3.1, so the reader has a working notion of the procedure before seeing it referenced in subsequent definitions:

We refer below and throughout the rest of this article to symbolic repair, the process of augmenting the symbolic transition set with new, dynamically feasible skills whenever the current specification becomes unrealizable. The full procedure is formalized in Sec. 5.3; for the purpose of the upcoming definitions, it suffices to treat symbolic repair as a black box that proposes candidate symbolic transitions whose physical feasibility is then verified by MICP.

For the Eq. (1) notation, we added an inline clarification just below the equation:

The function $\mathcal{D}(\cdot)$ encodes the discrete-time system dynamics that propagate the continuous state $\phi[i]$ to $\phi[i + 1]$; the specific dynamics used in this work are detailed in Eq. (3).

Response To Reviewer #5

Overall Comments

This paper pertains to a framework for motion planning, the crux of which is solving mixed-integer convex programs (MICPs). ... The paper adds some important content compared to the main point of reference, the IROS paper. First is of course the extensive (including real-world) experiments ... These experiments also include a benchmark in simulation, against pure MIP planning. ... Likely stemming from the need for real hardware experiments, the paper also adds a tracking control module based on nonlinear MPC.

Response

We thank the reviewer for the careful and constructive review, and in particular for noting the value of the tracking-control and benchmark additions. The clarity-of-presentation comments below all map to specific revisions in the IJRR submission, summarized item-by-item.

Reviewer Comment 1: “Reactive Synthesis” / LTL Not Defined

... where the introduction falters is that it doesn’t really, clearly define what the authors mean by “reactive synthesis” while employing it several times, along with “LTL”. ... I think the authors could more strongly direct the reader to read [bloem2012synthesis] before continuing with the rest of the paper, or direct them towards the reference cited in §III from the start ... The authors could, in the introduction still, even quickly describe what LTL is as they would to a layperson.

Response

We added an inline definitional paragraph in the introduction immediately after the first use of “LTL” / “reactive synthesis,” aimed at readers without a formal-methods background:

For readers unfamiliar with formal methods, LTL is a logic for expressing temporal properties of system behaviors using operators such as “always,” “eventually,” and “until” (Piternan et al., 2006; Bloem et al., 2012); reactive synthesis automatically constructs a controller (a strategy automaton) that satisfies an LTL specification against any admissible environment behavior, providing correct-by-construction guarantees at the symbolic level.

Reviewer Comment 2: Related-Work Section Reuses Jargon

... the related work section still suffers from the problem of the introduction, using the term “synthesis” and “Linear Temporal Logic (LTL)” as if the reader already grappled with these concepts. Another criticism I have of the related work section (§II) is that there is no mention at all of learning-based approaches ...

Response

With the introductory definitional paragraph in place (response to Comment 1), the LTL/synthesis usage in the related-work section now has an anchor for readers. We additionally added a paragraph on learning-based perceptive locomotion approaches and explicitly position our model-based formal-methods framework as complementary:

A complementary line of work uses end-to-end reinforcement learning to train perceptive locomotion policies that directly map terrain observations to joint commands, achieving impressive robustness on stairs, gaps, and unstructured outdoor terrain (Miki et al., 2022a; Agarwal et al., 2023; Zhuang et al., 2023; Hoeller et al., 2023; Cheng et al., 2024; Luo et al., 2024). Compared with these data-driven approaches, our model-based, formal-methods framework offers explicit symbolic-level realizability guarantees and physical-feasibility certificates from MICP, at the cost of weaker generalization and the need for a terrain abstraction; we view the two approaches as complementary rather than competing.

Reviewer Comment 3: MICP Solver

Which solver is used to solve the mixed-integer (convex) program? Is it Gurobi? CPLEX? Some other, homegrown MI(C)P solver? Have the authors compared how different solvers would perform?
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Response

We use Gurobi for all MICP problems in this work and have added an Implementation Details subsection at the start of the results section that states this explicitly. We did not benchmark across solvers; this is a reasonable additional study but is largely orthogonal to the symbolic-decomposition contribution we evaluate against the pure-MIP baseline (which uses the same solver in both configurations).

All MICP problems—including the offline gait-free MICP, the online gait-fixed MICP, the kinematic-feasibility re-targeting MICP, and the pure-MIP benchmark—are solved using Gurobi (Gurobi Optimization, LLC, 2025).

Reviewer Comment 4: Hardware (CPU, RAM)

What kind of hardware (outside of the robot – computer CPU, RAM) are the computations running on? AMD? Intel? Which architecture? How many cores?

Response

The Implementation Details subsection now reports the workstation specifications used for offline synthesis and online planning: an Intel Core i9-13900K (24 cores) with 64GB of RAM, running Gurobi 9.

Reviewer Comment 5: More Sophisticated Baseline

The authors compare with a heuristic on real hardware. What about another more sophisticated approach? Another hierarchical planner? A learning-based method?

Response

We appreciate this suggestion and agree that a stronger baseline would add value. While a full implementation of such a baseline was beyond the scope of this revision, we have (i) strengthened the rationale for the nearest-feasible-foothold heuristic in the hardware section, and (ii) added an explicit future-work commitment listing concrete stronger baselines—TOWR-style phase-based parameterization (Winkler et al., 2018) and parkour-style RL controllers (Zhuang et al., 2023; Hoeller et al., 2023; Cheng et al., 2024; Luo et al., 2024)—in the new *Stronger Baseline Comparisons as Future Work* subsection of the discussion (Sec. 10). The strengthened rationale in the hardware section reads:

We chose this nearest-feasible-foothold heuristic because it isolates the value of feasibility-aware, longer-horizon planning over a single-step heuristic, and because such heuristics remain widely used in deployed perceptive-locomotion stacks. Comparisons against more sophisticated baselines are discussed as future work in Sec. 10.

Reviewer Comment 6: Bipedal Extension

As in [zhou2025physically], this paper focuses on quadrupedal walking. Does this framework extend to bipedal robots? ... I think the conclusion would be appropriate for that.

Response

We added a dedicated subsection *Extension to Bipedal and Other Platforms* in the discussion (Sec. 10). The symbolic machinery is platform-agnostic; the MICP layer needs adaptation, and the angular-momentum simplification becomes a stronger limitation in the bipedal regime—these points are stated explicitly.

Reviewer Comment 7: Sections IV And V Are Very Short

Sections IV and V are very short. This is the most minor of minor comments but I feel the authors could rebalance a bit by merging them into a single section – “Problem statement and approach”?

Response

Done. The previously separate Problem Statement and Approach sections are now merged into a single section “Problem Statement and Approach Overview” (Sec. 4) with two subsections.

Reviewer Comment 8: Fig. 9 Error Bars / Table IV Std Deviation

Fig. 9, for the simulation benchmark ... is called a “histogram”. It’s rather a bar chart of the averages. I would like to see error bars/whiskers on here, to see the standard deviations. ... Table IV (solve times for the online gait-fixed MICP) would benefit from including the standard deviation for the solve times.

Response

We appreciate the suggestion and have addressed it on both fronts. Fig. 11 has been regenerated as **box plots** aggregated over 10 seeded waypoints per scenario, so the reader can directly read off the median, interquartile range, and full spread of per-MIP solve times for each (terrain, horizon) combination—rather than only the mean as in the prior bar-chart version. Table 6 (the hardware MICP solve-time table) also now includes a Std (s) column reporting the per-trial standard deviation for every scenario.

Reviewer Comment 9: Typos

“highly nonlinearity” (§II.B) ... “and timing are chosen *a priori*” (p. 3).

Response

Both typos are fixed: “high nonlinearity” and “*a priori*” in the related-work section.

Response To Reviewer #7

Overall Comments

The paper presents a hierarchical planning and control framework for legged locomotion on uneven terrain, extending the previously existing work [zhou2025physically]. ... The key contributions beyond [zhou2025physically] are: (1) a coordination mechanism between the symbolic planner and tracking controller ..., (2) online pose retargeting to ensure feasibility under actual terrain, and (3) comprehensive hardware validation with systematic benchmarking against both heuristic-based and pure MIP approaches.

Response

We thank the reviewer for the careful summary of the prior submission and for the detailed concerns raised. We address each below. The single most important change in this IJRR submission is that we have rebalanced the contribution list to give proper credit to the conference-introduced contributions (the integrated framework, the manager, and symbolic repair with dynamic feasibility checks), rather than leaning entirely on the journal extensions identified in the reviewer’s summary.

Reviewer Comment 1: Incremental Contributions And Scientific Novelty

The incremental contributions relative to [zhou2025physically] requires stronger justification. Hardware deployment and delay handling appear to be practical achievements with limited scientific contribution. The proposed coordination mechanism (waiting for MICP or appending trajectories) are straightforward implementation choices and do not appear to be innovative. The paper will benefit from a clearer identification and articulation of technical and scientific challenges that were overcome during hardware deployment.

Response

We thank the reviewer for raising this directly, and we have substantially revised the manuscript to make the journal contribution case clearer. We see the journal extensions as more than implementation choices. The delay-aware coordination scheme and the kinematic-feasibility re-targeting are the mechanisms that make the closed loop between the discrete strategy automaton and the continuous tracking controller well-defined under bounded MICP solve time and under terrain–pose mismatch at runtime—two issues that any deployable synthesis-with-MICP planner must resolve, and that no prior synthesis-MICP work had to address because no prior such work had been deployed end-to-end on real hardware. The collision-region selection with binary $\mathbf{H}_{s,w,j}$ variables is, similarly, a substantive change to the online MICP formulation rather than a re-implementation of the conference version. The introduction now contains a dedicated paragraph that motivates each of these mechanisms with explicit discussion of the concrete deployment gap it closes, presented alongside (and at the same level of detail as) the conference-introduced scalability machinery—the high-level manager and symbolic repair with dynamic feasibility checks—so that the framework’s

complete technical scope is visible end-to-end. The contribution list now reads (paraphrased from the introduction):

(1) *Integrated reactive-synthesis-and-MICP framework for terrain-adaptive locomotion.* (2) *Scalable offline synthesis via a high-level manager and symbolic repair with dynamic feasibility checks (80.9–90.1% state-space reduction; 71.6–97.6% MICP-solve reduction).* (3) *Online execution module that bridges offline synthesis and real-world deployment, with new mechanisms introduced in this article: kinematic-feasibility re-targeting, delay-aware coordination between the strategy automaton and the tracking controller, and collision-region selection in the online MICP.* (4) *Comprehensive empirical validation across simulation and hardware with two baselines.*

Beyond the contribution restructuring, we have also strengthened the *technical soundness* of the newly proposed mechanisms with new quantitative evidence. The Pure-MIP benchmark subsection (Sec. 8) now reports both (i) a per-MIP solve-time scaling study over 10 seeded waypoints per scenario and five planning horizons (Fig. 11) and (ii) a new end-to-end benchmark on the dense-rebar scenario (Table 5) with success rate, traversal time, and per-module computational cost. The same benchmark contains explicit *ablation rows* for the two newly proposed mechanisms—disabling the MIP feasibility check (FC) drops the proposed planner’s success rate from 90% to 60%, and disabling delay-aware coordination (DAC) inflates the mean traversal time from 18.5s to 23.4s—confirming that each mechanism is necessary for the reported performance. The per-reviewer responses on the following pages detail each change.

Reviewer Comment 2: LTL Spec Construction Details

Symbolic synthesis step: How are LTL specifications (ϕ) constructed from MICP solutions? Are preconditions and postconditions manually encoded or learned? Section VI-B states skills execute “when at least one of its preconditions is satisfied” – please clarify whether this refers to one precondition set ($\sigma_{\text{robot}}, \sigma_{\text{terrain}}$ pair) from Σ_o^{pre} , or individual propositions within a precondition.

Response

The two parts of the question are addressed in two places in the manuscript. First, at the start of the Task Specification Encoding subsection (Sec. 5.2), we now state explicitly that the precondition and postcondition sets of each skill are user-defined and that the GR(1) encoding is a deterministic function of those user-defined sets:

Each skill is associated with a user-defined precondition set $\Sigma_o^{\text{pre}} \subseteq 2^{\mathcal{I}_{\text{robot}} \cup \mathcal{I}_{\text{terrain}}}$ and postcondition set $\Sigma_o^{\text{post}} \subseteq 2^{\mathcal{I}_{\text{robot}}}$, expressed over the robot and terrain inputs introduced in Sec. 3.1. The encoding rules below map these sets into the GR(1) components $\varphi_e^{t, \text{skill}}$ and $\varphi_s^{t, \text{skill}}$ deterministically.

This clarifies that preconditions/postconditions are manually authored rather than learned, and that the LTL specification ϕ is constructed deterministically from them via the encoding rules in the same subsection. Second, the more specific question about the meaning of “a skill executes when at least one of its preconditions is satisfied” is addressed in place, immediately after the precondition equation defining $\varphi_s^{t, \text{skill}}$:

The disjunction $\bigvee_{\sigma \in \Sigma_o^{\text{pre}}}$ in the equation above is taken over precondition pairs in Σ_o^{pre} : a skill becomes eligible to execute when the conjunction of input assignments specified by at least one full pair $(\sigma_{\text{robot}}, \sigma_{\text{terrain}}) \in \Sigma_o^{\text{pre}}$ holds, rather than when only some individual atomic propositions within a pair hold.

Reviewer Comment 3: Calibrating Claims

Dynamically changing environments: All experiments use static, pre-mapped terrains. **Dynamic locomotion:** The hardware experiments primarily use conservative gaits (3s trotting, 2.25s static walking). **Safety guarantees:** The convex relaxations of dynamics . . . and lack of verification between MICP and NMPC layers limits formal safety assurances.

Response

We agree on all three points and have toned down the language throughout the paper:

(a) “**Dynamically changing environments**” is replaced with “unforeseen terrain configurations revealed at runtime” in the abstract and introduction. Runtime symbolic repair handles unexpected terrains revealed during execution; we do not claim continuous environmental change.

(b) “**Dynamic locomotion**” is replaced with “agile, variable-gait locomotion” or “terrain-adaptive locomotion” depending on context. We added a sentence in the hardware section calibrating the gait-timing claim:

The gait timings adopted on hardware—3 s trotting, 1.5 s leaping, 2.25 s static walking—are conservative compared to the most agile motions reported in the legged-locomotion literature; however, they exercise the framework’s gait-switching, online MICP, and runtime-repair mechanisms, which is the validation target of this article. Aggressive, high-speed gaits would require revisiting the convex-dynamics simplification.

(c) “**Safety guarantees**” is qualified as “symbolic-level realizability guarantees and physical-feasibility certificates from MICP.” We further added an explicit angular-momentum disclosure paragraph after the dynamics equation in the offline synthesis section, and the discussion section now flags the lack of formal MICP/NMPC verification as future work.

Reviewer Comment 4: Limited Reactivity / Classical Sense-Plan-Act

The symbolic-MICP level interaction operates at a low frequency (once per MICP solve, $\sim 0.37\text{--}9.13\text{s}$), with no symbolic/MICP level involvement during NMPC tracking. . . . This resembles a classical sense-plan-act cycle more than reactive synthesis. How does this differ from a single MICP controller . . . ? Could you demonstrate scenarios where symbolic-level reasoning provides qualitative advantages . . . ? What prevents more frequent symbolic-continuous interaction . . . ?

Response

We address the three sub-questions in turn.

What prevents more frequent symbolic–continuous interaction? Two computational bottlenecks. (i) On the symbolic side, reactive synthesis has exponential time complexity in the number of variables in the specification (Bloem et al., 2012); running the symbolic layer at higher frequency would generally require a finer abstraction (more atomic propositions per cell, finer state granularity, a larger skill repertoire), which inflates the synthesis cost rapidly. (ii) On the MICP side, each per-transition online gait-fixed MICP itself takes hundreds of milliseconds to several seconds (Tables 4 and 6), which precludes running it in a true predictive fashion at the NMPC tick rate. Crucially, neither limit is architectural in the proposed framework: the delay-aware coordination

scheme already supports asynchronous MICP solves of arbitrary duration, so any solver speed-up or any careful refinement of the abstraction translates directly into higher update rates. We have added a new *Reactivity and Update Frequencies* subsection in the discussion (Sec. 10) that lays out both bottlenecks and the available paths to higher reactivity:

More frequent MICP updates are easily adaptable in the proposed framework: the delay-aware coordination scheme already supports asynchronous MICP solves of arbitrary duration, so any speed-up of the underlying solver translates directly into a higher MICP update rate. . . . More frequent symbolic-level planning is also achievable but requires careful design of the symbolic abstraction. Higher-frequency symbolic decisions would generally call for a finer abstraction . . . Reactive synthesis, however, scales exponentially in the number of variables in the specification, so any finer abstraction must be balanced against synthesis cost.

Where does our framework’s complexity sit relative to a single MICP controller, and where does symbolic reasoning provide qualitative advantage? The pure-MIP benchmark in Fig. 11 is the natural baseline—it is exactly a single MICP controller without symbolic decomposition. To anchor the comparison: the 2s pure-MIP horizon is configured with a planning region of two cells (i.e., two consecutive symbolic transitions in our framework), with each additional second of horizon adding half a cell. The 2s column therefore reads as a like-for-like comparison for an equivalent two-transition footprint, with the longer-horizon columns quantifying the cost of folding a larger region into a single MICP solve. Compared with the per-transition online gait-fixed MICP solve times in Table 4, our framework sits at the cheap end of the same complexity continuum, with the symbolic layer adding long-horizon reasoning at negligible per-transition overhead.

The qualitative advantage of the symbolic layer is now also confirmed quantitatively by the new end-to-end benchmark on the dense-rebar scenario (Table 5). Against 10 identically seeded random waypoints, the proposed planner attains 90% success at 18.5s mean traversal time and 5.26s mean MIP transition cost per trial; the pure-MIP $H=3$ s baseline (the natural single-MICP-controller comparison) matches the 90% success rate but at 24.7s traversal and 11.54s MIP cost, and the shorter $H=2$ s pure-MIP horizon drops to 60%. Fig. 12 shows a representative trial in which the short-horizon pure-MIP planner stalls in a local minimum—it has no mechanism to deliberate at the symbolic level about which neighboring cell to attempt next—whereas the proposed planner routes through a different sequence of cells. The same qualitative argument carries over to short-horizon receding-horizon MIP baselines regardless of solve speed (see also Reviewer 4’s Comment 8).

How does this differ from a single MICP controller with gait-sequencing heuristics and a gait-free fallback? We read this as referring to a baseline that, at every step, iterates over a candidate set of gaits, solves a gait-fixed MICP for each candidate, and falls back to a gait-free MICP whenever the candidate gaits all fail. We see two key differences from such a baseline.

First, designing a gait-sequencing heuristic that generalizes is itself non-trivial. The set of feasible gait choices depends on the local terrain in ways that are not captured by simple rules—e.g., a 1-second trotting gait is feasible on a flat region but not on a wide gap; a 1.5-second leaping gait is feasible across a gap but unnecessarily aggressive on a flat-and-dense region; *extreme sparse* rebar configurations admit only specific footstep patterns. The hand-tuning required to enumerate which gait to try first across these mixed-terrain scenarios—and to define the iteration order for fallbacks—is precisely the engineering burden that hierarchical formal-methods planners are designed to remove.

Second, our offline synthesis already encodes per-skill feasibility (verified by gait-fixed MICP, with gait-free MICP only invoked during repair), and the synthesized strategy automaton uses this information to select a gait whose feasibility on the current symbolic abstraction is already certified. At runtime, this typically saves multiple online MICP solves per step that an iterate-and-test heuristic would incur: in the iterate-and-test setting, a wrong-gait guess pays the full online

MICP cost before being detected; if all candidate gaits fail, an additional gait-free MICP must be solved (which is the most expensive call in our pipeline—averaging 27.23 s in our experiments). Our framework collapses this iteration into a single online gait-fixed MICP per transition under nominal conditions, with online repair (and only then a gait-free MICP) reserved for genuinely unforeseen terrains.

Reviewer Comment 5: Heuristic Baseline + Pure MIP Definition + Ablation

(a) **Heuristic baseline:** ...overly simplistic. Consider comparing against recent perceptive locomotion methods such as Winkler 2018 ... (b) **Pure MIP baseline:** requires major clarifications. (c) **Ablation studies:** Analyze contributions of (1) online retargeting, (2) delay coordination, (3) symbolic vs. direct MICP replanning.

Response

(a) **Heuristic baseline.** We appreciate the suggestion. While we did not have the bandwidth to implement a more sophisticated baseline within the scope of this revision, we have strengthened the rationale for the nearest-feasible-foothold heuristic in the hardware section, and added an explicit future-work commitment to concrete stronger baselines in the discussion. The strengthened rationale reads:

We chose this nearest-feasible-foothold heuristic because it isolates the value of feasibility-aware, longer-horizon planning over a single-step heuristic, and because such heuristics remain widely used in deployed perceptive-locomotion stacks. Comparisons against more sophisticated baselines are discussed as future work in Sec. 10.

The new *Stronger Baseline Comparisons as Future Work* subsection of the discussion section lists Winkler 2018 (Winkler et al., 2018) and parkour-style RL controllers (Zhuang et al., 2023; Hoeller et al., 2023; Cheng et al., 2024; Luo et al., 2024) as concrete future-baseline targets, and notes that a fair comparison would require either equipping these baselines with a comparable symbolic decomposition layer or restricting our framework to single-skill operation.

(b) **Pure MIP baseline.** The benchmark subsection now precisely defines the pure-MIP baseline:

The pure-MIP baseline uses the same convex relaxations of the dynamics, the same linearized friction-cone and fixed-Jacobian kinematic constraints, and the same cost weights as our online gait-fixed MICP. The only differences are (i) the planner solves a single long-horizon gait-fixed MICP without symbolic-level decomposition, and (ii) the local planning region is expanded to encompass all terrain polygons within the horizon-induced reachable distance. The same Gurobi solver and settings are used in both cases.

(c) **Ablation studies.** We have added explicit ablation experiments for the two newly proposed online-execution mechanisms—the **MIP feasibility check (FC)**, which corresponds to the kinematic-feasibility re-targeting step, and the **delay-aware coordination (DAC)** scheme—as part of a new end-to-end benchmark on the dense-rebar scenario (Sec. 8, Table 5). Each method is dispatched against 10 identically seeded random waypoints, with rebars randomly removed per trial to add difficulty. The condensed headline numbers are:

Method	Success	Traversal (s)	MIP trans. (s)
Proposed	90%	18.5	5.3
no FC	60%	20.1	5.9
no DAC	90%	23.4	5.1
Pure-MIP ($H=3$ s)	90%	24.7	11.5
Pure-MIP ($H=2$ s)	60%	14.8	3.0
Pure-MIP ($H=2$ s), no FC	20%	9.0	1.4

The ablation rows confirm the value of both newly proposed mechanisms:

- Disabling the FC drops the proposed planner from 90% to 60% success and the pure-MIP $H=2$ s baseline from 60% to 20%, because the planner can no longer prune symbolic transitions whose target poses are not physically reachable on the perceived terrain.
- Disabling DAC preserves the 90% success rate but inflates the mean traversal time from 18.5 s to 23.4 s, because each symbolic transition now waits for the previous trajectory to be fully tracked before the next MIP can start.

For the third question (symbolic vs. direct MICP replanning), the pure-MIP rows in the same table serve as a direct ablation of the symbolic decomposition: the proposed planner matches the pure-MIP $H=3$ s success rate (90%) at roughly three quarters of the traversal time and less than half of the MIP-transition cost per trial; the shorter $H=2$ s pure-MIP horizon drops to 60%, so without symbolic guidance the horizon must be expanded (and the MIP cost incurred) for the planner to recover from the harder rebar gaps. Collision-region selection is separately exercised by the *Unstructured-8* simulation scenario (Table 4), where activating it raises the binary-variable count from 69 to 256 and the solve time correspondingly by 440%; we did not run a standalone disable-collision-region ablation because the mechanism is not optional for that scenario.

Reviewer Comment 6: Section Organization

Please consider reorganizing sections better. For example, Section IV is just seven lines long.

Response

The previously short Approach section has been merged into a single “Problem Statement and Approach Overview” section (Sec. 4), aligned with Reviewer 5’s analogous suggestion.

Reviewer Comment 7: Perception Validation / Uncertainty

While focusing on planning/control is reasonable, the lack of online perception limits validation of safety claims. Consider discussing how terrain segmentation uncertainty or perception delays would affect the framework.

Response

The discussion section now includes a paragraph on perception uncertainty, segmentation noise, polygon misalignment, and the partial fallback that runtime symbolic repair provides. We also added an explicit out-of-scope statement on perception in the online execution section.

Reviewer Comment 8: Rationale For Two-Level Discrete Decisions

The rationale for making discrete decisions at both symbolic-level and MICP-level could be clarified. What does the symbolic level provide that a better-designed MICP level couldn't achieve? Or alternately, why can't your symbolic level decide which polygon to step on?

Response

Given that the MICP already makes binary decisions (contact mode, region selection), one may ask why a separate symbolic decomposition is needed. The short answer is the state-space blowup. Pushing region-selection decisions for an entire local environment into the symbolic layer would require enumerating all combinations of polygon assignments and gait choices over the synthesis state, which scales poorly in both polygon count and horizon. Conversely, expanding the MICP to cover multi-cell horizons gives back the long-horizon scaling we set out to avoid (Fig. 11). The symbolic layer therefore reasons over a coarse abstraction (cell-level robot states, terrain-type categories, gait identifiers) that admits formal realizability guarantees, while the MICP handles fine-grained continuous and combinatorial decisions inside a single transition. The two layers together cover the long-horizon decision-making space at a tractable computational budget. We chose not to fold this rationale into the manuscript itself because it is essentially a defense of an existing design choice rather than new content.

References

- Ananye Agarwal, Ashish Kumar, Jitendra Malik, and Deepak Pathak. Legged locomotion in challenging terrains using egocentric vision. In *Conference on robot learning*, pages 403–415. PMLR, 2023.
- Roderick Bloem, Barbara Jobstmann, Nir Piterman, Amir Pnueli, and Yaniv Saar. Synthesis of reactive (1) designs. *Journal of Computer and System Sciences*, 78(3):911–938, 2012.
- Xuxin Cheng, Kexin Shi, Ananye Agarwal, and Deepak Pathak. Extreme parkour with legged robots. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 11443–11450. IEEE, 2024.
- Péter Fankhauser, Marko Bjelonic, C Dario Bellicoso, Takahiro Miki, and Marco Hutter. Robust rough-terrain locomotion with a quadrupedal robot. In *2018 IEEE International Conference on Robotics and Automation (ICRA)*, pages 5761–5768. IEEE, 2018.
- Gurobi Optimization, LLC. Gurobi Optimizer Reference Manual. <https://www.gurobi.com>, 2025.
- David Hoeller, Nikita Rudin, Dhionis Sako, and Marco Hutter. Anymal parkour: Learning agile navigation for quadrupedal robots. *arXiv preprint arXiv:2306.14874*, 2023.

- Shixin Luo, Songbo Li, Ruiqi Yu, Zhicheng Wang, Jun Wu, and Qiuguo Zhu. Pie: Parkour with implicit-explicit learning framework for legged robots. *IEEE Robotics and Automation Letters*, 9(11):9986–9993, 2024.
- Takahiro Miki, Joonho Lee, Jemin Hwangbo, Lorenz Wellhausen, Vladlen Koltun, and Marco Hutter. Learning robust perceptive locomotion for quadrupedal robots in the wild. *Science robotics*, 7(62):eabk2822, 2022a.
- Takahiro Miki, Lorenz Wellhausen, Ruben Grandia, Fabian Jenelten, Timon Homberger, and Marco Hutter. Elevation mapping for locomotion and navigation using gpu. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 2273–2280. IEEE, 2022b.
- Nir Piterman, Amir Pnueli, and Yaniv Sa’ar. Synthesis of reactive (1) designs. In *Verification, Model Checking, and Abstract Interpretation: 7th International Conference, VMCAI 2006, Charleston, SC, USA, January 8-10, 2006. Proceedings 7*, pages 364–380. Springer, 2006.
- Alexander W Winkler, C Dario Bellicoso, Marco Hutter, and Jonas Buchli. Gait and trajectory optimization for legged systems through phase-based end-effector parameterization. *IEEE Robotics and Automation Letters*, 3(3):1560–1567, 2018.
- Ziwen Zhuang, Zipeng Fu, Jianren Wang, Christopher Atkeson, Soeren Schwertfeger, Chelsea Finn, and Hang Zhao. Robot parkour learning. *arXiv preprint arXiv:2309.05665*, 2023.